

Tutorial group: _____

Name: _____

Matriculation Number: _____

September 2025

MH1812 Test 1

60 minutes

QUESTION 1.

(30 marks)

Use mathematical induction to show that

$$\frac{1}{4 \cdot 9} + \frac{1}{9 \cdot 14} + \cdots + \frac{1}{(5n-1)(5n+4)} = \frac{n}{4(5n+4)}.$$

for all integers $n \geq 1$.

[Proof:]

Base case: when $n = 1$, the left-hand side is $1/(4 \cdot 9)$ and the right-hand side is $1/(4 \cdot (5+4)) = 1/36$. The equality holds.

Inductive step: Suppose that the equality holds for n , and we shall show that it holds for $n+1$, that is, we shall show that

$$\frac{1}{4 \cdot 9} + \frac{1}{9 \cdot 14} + \cdots + \frac{1}{(5n-1)(5n+4)} + \frac{1}{(5n+4)(5n+9)} = \frac{n+1}{4(5n+9)}. \quad (1)$$

By induction hypothesis, the LHS of (1) equals

$$\frac{n}{4(5n+4)} + \frac{1}{(5n+4)(5n+9)} = \frac{n(5n+9) + 4}{4(5n+4)(5n+9)} = \frac{(5n+4)(n+1)}{4(5n+4)(5n+9)} = \frac{n+1}{4(5n+9)},$$

which is exactly the RHS of (1).

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 1$.

QUESTION 2.

(30 marks)

(a) (10 marks) Find $12012^{2025} \bmod 15$.

[Solution 1:] Since $12012 \equiv 12 \equiv -3 \pmod{15}$, we just need to find $(-3)^{2025} \bmod 15 = -3^{2025} \bmod 15$.

We argue that $3^{n+4} \equiv 3^n \pmod{15}$ for $n \geq 1$. Indeed, $3^{n+4} - 3^n = 3^n(3^4 - 1) = 3^n \cdot 80$, and $3 \mid 3^n$, $5 \mid 80$, thus $15 \mid (3^n \cdot 80)$, i.e., $15 \mid (3^{n+4} - 3^n)$. This proves the claim. It follows that

$$-3^{2025} \equiv -3^{2025 \bmod 4} \equiv -3 \pmod{15}.$$

The answer is therefore $12012^{2025} \bmod 15 = 12$.

[Solution 2:] Since $12012 \bmod 15 = 12$, we shall find $r = 12^{2025} \bmod 15$. Write $12^{2025} = 15q + r$, we know that $3 \mid r$, and dividing both sides by 3, we obtain that

$$4 \cdot 12^{2024} = 5q + (r/3).$$

We then find $12^{2024} \bmod 5$. Note that $12^{2024} \equiv 2^{2024} \equiv 4^{1012} \equiv (-1)^{1012} \equiv 1 \pmod{5}$, we see that $12^{2024} \bmod 5 = 1$. Thus, $r/3 = 4$ and $r = 12$. Therefore, $12012^{2025} \bmod 15 = 12$.

(b) Let $\mathbb{Z}^+$ denote the set of positive integers. For $x, y \in \mathbb{Z}^+$, let $P(x, y)$ denote the predicate “ $4 \mid (x + y + 1)$ ”. What are the truth values of these statements?

(i) (10 marks) $\forall x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+, P(x, y)$.

(ii) (10 marks) $\exists x \in \mathbb{Z}^+, \forall y \in \mathbb{Z}^+, P(x, y)$.

[Solution:]

(i) The statement is true. For any given x , take $y = 4 - ((x + 1) \bmod 4)$, then $4 \mid (x + y + 1)$.

(ii) The statement is false. For any given x , take $y = 5 - ((x + 1) \bmod 4)$, then $4 \nmid (x + y + 1)$.

QUESTION 3.**(40 marks)**

(a) (20 marks) Determine whether $((p \vee q) \rightarrow r) \vee (p \wedge (r \rightarrow q))$ is a tautology.

[Proof:] It is not a tautology. One can take $p \equiv F$, $q \equiv T$ and $r \equiv F$, the expression evaluates to $(T \rightarrow F) \vee F \equiv F$.

(b) (20 marks) Determine whether the following argument is valid. (State the names of the inference rules whenever you apply them.¹)

$$\begin{aligned} & (r \wedge t) \rightarrow u; \\ & p \rightarrow t; \\ & p \vee r; \\ \therefore & u. \end{aligned}$$

[Solution:] The argument is invalid. When $p \equiv F$, $r \equiv T$, $t \equiv F$, $u \equiv F$, all premises are true while the conclusion is false.

¹The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, hypothetical syllogism and rule of contradiction.

BONUS QUESTION.**(10 marks)**

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

Find all integral solutions to $x^2 + y^2 + z^2 = x^2y^2$.

[**Solution 1:**] The answer is the zero solution only.

To see this, we take both sides modulo 4. Note that $n^2 \equiv 0, 1 \pmod{4}$ for all integers n . If the right-hand side is 1, we see that $x^2 \equiv y^2 \equiv 1$ and thus $x^2 + y^2 + z^2 \equiv 2, 3 \pmod{4}$ and we have a contradiction. Thus the right-hand side must be 0 and we have $x^2 + y^2 + z^2 \equiv 0 \pmod{4}$ and it must hold that $x^2 \equiv y^2 \equiv z^2 \equiv 0 \pmod{4}$. Therefore, x , y and z are all even and we can write $x = 2x'$, $y = 2y'$ and $z = 2z'$. It then follows that

$$(x')^2 + (y')^2 + (z')^2 = 4(x')^2(y')^2.$$

This means that

$$(x')^2 + (y')^2 + (z')^2 \equiv 0 \pmod{4}.$$

If x , y and z are not all zeroes and $4|(x^2 + y^2 + z^2)$, then we have another tuple (x, y, z) satisfying $4|(x^2 + y^2 + z^2)$ with a strictly smaller value of $x^2 + y^2 + z^2$. Such process cannot continue indefinitely since all variables are integers. (More rigorously, we can pick a nonzero solution with the smallest value of $x^2 + y^2 + z^2$ and obtain a contradiction.) Therefore, there cannot exist a nonzero solution.

[**Solution 2:**] The given equation can be transformed as

$$z^2 + 1 = (x^2 - 1)(y^2 - 1).$$

The left-hand side is never an integer multiple of 4 and thus the right-hand side is not an integer multiple of 4, either. This means that both x and y are even, then z must also be even. We can then proceed as in Solution 1 using an infinite descent argument.

[**Solution 3:**] In a nonzero solution, we must have $x, y \neq 0$. As in the previous solution, we transform the given equation into

$$z^2 + 1 = (x^2 - 1)(y^2 - 1)$$

and make the observation that x and y must be even.

Without loss of generality, suppose that $x \neq 0$ in a nonzero solution. Since x is even, one of $x - 1$ and $x + 1$ must be congruent to 1 modulo 4 and the other to 3 modulo 4. Therefore, the right-hand side is divisible by some prime $p \equiv 3 \pmod{4}$.

Now we argue that each prime $p \equiv 3 \pmod{4}$ cannot be a factor of the left-hand side, we then meet a contradiction and can conclude that $(0, 0, 0)$ is the only integral solution. (This claim is related to a theory called *quadratic reciprocity*, but here we just present need a simple proof.) Suppose that $p \mid (z^2 + 1)$, where p is a prime and $p \equiv 3 \pmod{4}$. Then $(p - 1)/2$ is an odd integer and

$$z^{p-1} \equiv (z^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} \equiv -1 \pmod{p}$$

Since $p \mid (z^2 + 1)$, we know that $p \nmid z$. By Fermat's Little Theorem, we know that $z^{p-1} \equiv 1 \pmod{p}$. We have met a contradiction and thus proven the claim.